1. INTRODUCTION

In the modern digital world, cybersecurity plays a critical role in protecting sensitive information and maintaining system integrity. Organizations rely on security tools such as firewalls and secure communication protocols to defend against cyber threats. However, attackers continuously evolve their techniques to bypass these defenses.

This project focuses on two important case studies:

- Firewall failure despite implementation
- Protocol-level attack due to HTTP vulnerabilities

The goal is to analyze these failures and suggest effective modern security solutions.

2. SECTION – 1: FIREWALL AND NETWORK ARCHITECTURE CASE STUDY

2.1 Case Study Overview

A company experienced a major data breach despite having a firewall deployed in its network. Attackers were able to bypass the firewall and gain unauthorized access to sensitive customer data. The breach occurred through seemingly legitimate network traffic.

2.2 Types of Firewalls and Their Working

1. Packet Filtering Firewall

- Operates at Network Layer
- Filters packets based on:

➤ IP Address

➤ Port Number

➤ Protocol

2. Stateful Firewall

- Maintains a table of active connections
- Allows packets belonging to trusted sessions

Tracks session state

2.3 Analysis of Firewall Failure

Packet Filtering Firewall Failure

- Cannot inspect packet payload
- Allows malicious traffic through open ports (80/443)
- No understanding of application-layer attacks

Stateful Firewall Failure

- Trusts already established connections
- Cannot detect malicious activity within sessions
- Limited visibility into encrypted HTTPS traffic

2.4 Root Cause of Breach

- Lack of Deep Packet Inspection (DPI)
- No application-layer security
- No inspection of encrypted traffic
- Over-reliance on traditional firewall

2.5 Recommended Security Solutions

1. Next-Generation Firewall (NGFW)

- Deep Packet Inspection
- Intrusion Prevention System (IPS)
- Application awareness
- Real-time threat detection

2. Web Application Firewall (WAF)

- Protects web applications
- Prevents:
- SQL Injection
- Cross-Site Scripting (XSS)
- Session hijacking

3. Additional Security Measures

- Intrusion Detection/Prevention Systems (IDS/IPS)
- Endpoint Detection and Response (EDR)
- Secure Web Gateway (SWG)
- Network segmentation

SSL/TLS inspection

2.6 Secure Network Architecture Diagram

Copy code

```
Internet
  ↓
NGFW Firewall
  ↓
Web Application Firewall (WAF)
  ↓
Application Server
  ↓
Database Server
```

3. SECTION – 2: PROTOCOL ANALYSIS CASE STUDY

3.1 Case Study Overview

A web-based system was compromised due to:

- HTTP header manipulation
- Lack of HSTS (HTTP Strict Transport Security)

Attackers intercepted communication and manipulated data between users and servers.

3.2 Attack Analysis

1. HTTP Header Manipulation

Attackers modify HTTP headers such as:

- Cookie
- Host
- Authorization

Impact

- Session hijacking
- Identity spoofing
- Unauthorized access

2. Lack of HSTS

HSTS ensures browsers use HTTPS only.

Without HSTS:

- Users connect via HTTP
- Attackers perform:
- ➤ Man-in-the-Middle (MITM) attacks
- ➤ SSL stripping

3.3 Root Cause of Attack

- No HTTPS enforcement
- Weak validation of headers
- Poor session management
- Lack of secure configuration

3.4 Preventive Measures

1. Enable HSTS

- Forces HTTPS connections
- Prevents downgrade attacks

2. Use HTTPS Everywhere

- Redirect HTTP to HTTPS
- Install SSL/TLS certificates

3. Secure HTTP Headers

Use:

- HttpOnly cookies
- Secure cookies

4. Input Validation

- Sanitize all inputs
- Prevent injection attacks

5. Deploy Web Application Firewall (WAF)

- Detect malicious requests
- Block header manipulation

6. Strong Session Management

- Secure session IDs
- Session expiration
- Avoid storing sensitive data in headers

3.5 Attack Flow Diagram

User → HTTP Request → Attacker (MITM) → Modified Request → Server

4. CONCLUSION

This project demonstrates that traditional firewalls alone are insufficient to protect against modern cyber threats. Attackers exploit weaknesses in both network infrastructure and communication protocols.

To ensure strong security, organizations must adopt a multi-layered defense approach,

including:

- Next-Generation Firewalls
- Web Application Firewalls
- Secure communication protocols (HTTPS + HSTS)
- Continuous monitoring systems

A proactive security strategy is essential for preventing future cyber attacks.